INSIGHTSTACK

Enterprise SaaS Analytics Platform Source-Code Audit Report

Audit findings and prioritized implementation plan

Prepared 18 October 2026

Scope: React/TypeScript frontend, Django REST backend, routes, views, serializers, models, migrations, settings, dependencies, and database schema. The audit found substantial mock/local-only UI behavior and missing project database tables.

Audit only ? no project files were modified. This report documents findings and recommended work; implementation requires approval.

Executive summary

The application has multiple Django REST endpoints and domain models, but the frontend feature pages generally do not call the available data services. Login and registration simulate successful authentication with demo tokens, analytics and dashboard values are fixtures, and important buttons have no handlers. The existing SQLite database contains framework tables but no InsightStack domain-model tables. Report generation is particularly incomplete: the report page is mock-only, and the backend export endpoint returns placeholder content.

Validation evidence

Django system check passed. SQLite inspection found no project application tables. No project migration packages were present; a dry-run for the users app showed an initial migration would be generated. No source files were changed and no tests were run.

Priority key: P0 = blocks core usage or persistence; P1 = core product functionality; P2 = supporting/administrative capability.

A. Feature-by-feature status

Feature	Frontend status	Backend status	Database integration	Main issue	Priority
Registration, login, JWT	Demo-only; forms do not call APIs.	Register/login/refresh/logout routes exist; reset/email/MFA verification are placeholders.	User/profile models; app tables absent from SQLite.	Demo tokens cannot authenticate; no tenant onboarding after registration.	P0
Dashboard KPIs, charts, filters	Mock fixtures; range buttons slice fixtures.	KPI CRUD; revenue/customer data random; trend/conversion fixed.	Models exist; no app tables; snapshots not populated/exposed.	No API data load; metrics not derived from uploads.	P1
Dataset upload and preview	UI only; file input/drop not wired; delete local only.	CSV/XLSX parser endpoint; JSON unsupported.	Dataset stores file and preview metadata, not queryable records; tables absent.	Upload handler absent; service omits file_type; records/history routes missing.	P0
Revenue/customer/traffic analytics	Charts and summaries are mock/literal.	Revenue/customer endpoints random; traffic/regional/session missing.	No fact pipeline or app tables.	Service route mismatches; page never calls API.	P1
AI natural-language analytics	Hard-coded chat response and fixture insights.	Gemini routes exist; query requires workspace_id omitted by service; insights use mock KPI context.	Conversation/message models; tables absent.	UI not wired; backend context is not actual workspace data.	P1
Anomaly detection/forecast	Mock insight cards.	Helper exists; no view/task/endpoint for detection or forecast.	No result model; KPI/snapshots unused.	No implemented workflow.	P1
Reports and scheduling	Mock list; New/Download/Share controls have no handlers.	Report/schedule CRUD and share exist; exports are placeholders; no task generation.	Models exist; app tables absent.	No generation flow; export path mismatch; no scheduled delivery.	P0
PDF/Excel exports	No working download actions.	PDF and CSV only, placeholder content; no Excel.	No artifact storage.	Service uses a different URL shape; Excel unsupported.	P1
User/organization management	Fixture member list; local-only remove; invite/save inert.	Organization/workspace/member CRUD exists; invitation route absent.	Models exist; app tables absent.	Invite service targets missing route; nested membership authorization gaps.	P1

Feature	Frontend status	Backend status	Database integration	Main issue	Priority
Notifications	Fixture data; read actions local-only.	CRUD, read and unread actions exist.	Models exist; app tables absent.	PATCH/mark-all client routes mismatch backend POST/read-all.	P1
Audit logs	Fixture-only; filter local.	Read endpoints exist.	Models exist; app tables absent.	No audit/activity writers found.	P2
Subscriptions/admin	Hard-coded stats/plans/usage; controls inert.	Plan/current/upgrade endpoints exist.	Models exist; app tables absent.	No seed plans/payment; upgrade only edits a record; admin UI not role-gated.	P2
Dashboards/builder	Mock list; builder state only; save/preview inert.	Dashboard/widget CRUD exists.	Models exist; app tables absent.	New Dashboard route missing; widget API path mismatch.	P1
Profile/security/preferences	Save/photo/MFA controls inert; theme local.	Profile update and MFA setup exist; verification placeholder.	User profile model; app tables absent.	Settings page does not use API services.	P2

B. Mock and hard-coded data sources

Frontend fixture hub: frontend/src/utils/mockData.ts defines mock KPIs, revenue, activity, dashboards, customer growth, traffic, regions, monthly summaries, AI insights, reports, datasets, notifications, audit logs, team members, and plans. Replace each consumer with authenticated, workspace-scoped API results derived from persisted data.

Source file	Mock/hard-coded content	Expected real replacement
frontend/src/pages/app/DashboardHome.tsx	KPI cards, revenue charts, activity feed; ?Welcome back, Alex? and live timestamp.	Current-user endpoint; KPI and analytics aggregation endpoints; real audit/activity feed.
frontend/src/pages/app/Analytics.tsx	Revenue/customer/traffic/regional charts; literal summary cards and monthly sessions.	Workspace- and period-filtered analytics built from normalized uploaded records and real dimensions.
frontend/src/pages/app/AIInsights.tsx	Fabricated chat answer; fixture insight cards.	Gemini query with verified workspace context; persisted/generated insights from real metrics.
frontend/src/pages/app/DashboardBuilder.tsx	Literal $2.4M KPI and fixture-driven widget charts.	Configured KPI/dataset-backed widget queries, persisted with dashboard/widget APIs.
frontend/src/pages/app/DashboardList.tsx	Mock dashboards and counts.	Dashboard list/detail APIs scoped to selected workspace.
frontend/src/pages/app/Reports.tsx	Fixture reports and statuses.	Report CRUD/list and generated-artifact status APIs.
frontend/src/pages/app/Datasets.tsx	Fixture datasets, counts, sizes, and status.	Dataset list/upload APIs; real persisted metadata and preview/records.
frontend/src/pages/app/Organization.tsx	Fixture members; literal Acme Corp settings.	Organization/member API scoped to selected org, with authorization.
frontend/src/pages/app/AdminPanel.tsx	Static usage/stats and fixture plans/members.	Usage metrics, subscription plan, organization memberships APIs; restrict admin data.
frontend/src/pages/app/Notifications.tsx; components/layout/NotificationPanel.tsx	Fixture notifications and component-local read state.	Notification list/read/unread endpoints.
frontend/src/pages/app/AuditLogs.tsx	Fixture audit events.	Audit endpoint backed by event writers.
frontend/src/components/layout/Topbar.tsx	Fixed unread badge ?3?.	Notification unread-count endpoint.
backend/analytics/views.py	Random revenue/customer series, fixed trend and conversion values.	Aggregate persisted workspace records with requested time period and explicit empty/error handling.
backend/ai_insights/views.py	Mock KPI context.	Query authorized workspace KPI/dataset data before composing Gemini context.
backend/reports/views.py	Literal mock PDF and CSV payloads.	Render report configuration and underlying data to valid PDF/XLSX/CSV; store or stream generated artifact.

C. Report-generation failure trace

1. Reports.tsx initializes its list from mockReports. New Report, Download, and Share buttons have no onClick handlers. No frontend request is made.

2. services/index.ts has reportService.create() for POST /api/reports/, but Reports.tsx never imports or calls it. This endpoint creates report metadata; it is not report generation.

3. reports/urls.py registers ReportViewSet CRUD. The backend export action is GET /api/reports/{id}/export/?format=pdf (or csv). The frontend service instead calls /api/reports/{id}/export/pdf/ and similar paths, so its URL does not match the action route.

4. Even with a corrected URL, reports/views.py writes placeholder PDF bytes or fixed CSV text. It does not render the report or its underlying data. Excel is not supported. Schedules have no generation/delivery task.

5. UI fixture IDs are simple strings while backend IDs are UUIDs; fixture rows cannot be used as real report identifiers.

Conclusion: visible report controls are disconnected; the API contract and export format are mismatched; backend generation is stubbed; scheduled reports do not execute.

D. Missing endpoints, persistence, security, configuration

Migrations and database

No project-app migration packages were found. showmigrations listed only framework migrations; read-only SQLite schema inspection found framework tables and no InsightStack domain tables. A users migration dry-run indicated an initial migration is needed. Project models are loaded by Django, but model definitions alone do not create database schema.

Missing/mismatched API behavior

Area	Observed gap
Datasets	No records or history endpoints although frontend service requests both; no normalized imported-row model; JSON parsing unsupported; upload form omits file_type.
Analytics	Frontend customers/conversions/snapshots URLs disagree with backend customer-growth/conversion-rate routes; snapshots, traffic, regional, and session endpoints missing.
Dashboards	Frontend nests widgets under dashboard ID, while backend registers widgets at router root.
Organizations	Frontend calls /members/invite/, which does not exist; no email invitation/token workflow.
Notifications	Frontend PATCH /read/ and POST /mark-all-read/ disagree with backend POST /read/ and POST /read-all/.
Reports	No generation job/endpoint, Excel exporter, real PDF generation, artifact workflow, or scheduled delivery task.
Anomaly/forecast	No view/task/endpoint or persisted results; detect_trends helper is not invoked.
Audit	Read endpoints/models exist, but no audit/activity writers were found.

Authentication, authorization, tenant isolation

Frontend route protection trusts persisted client state; current login/register bypass backend entirely. Backend workspace-scoped list querysets exist in several viewsets, but create serializers do not consistently validate workspace membership. Nested organization/workspace member queries are scoped by URL ID, not by requester membership/role. User list/detail endpoints are available to any authenticated user. Admin UI route has no role check. Registration creates a user/profile but not organization/workspace.

Configuration and dependency gaps

base.py reads DATABASE_URL while .env.example lists DB_NAME/DB_USER/DB_PASSWORD/DB_HOST/DB_PORT. The actual backend/.env contents were not accessed or included. Django config references django_redis, absent from requirements.txt (development overrides the cache). No email sender/backend settings were found for reset, invitation, or scheduled email workflows. Media upload settings are absent. GEMINI_API_KEY is read but defaults empty. README describes React 18, while package.json declares React 19 and React 19 types.

E. Prioritized implementation plan

Order	Implementation	Why / verification outcome
1 ? P0	Generate/apply initial migrations for every project app; align DB env naming; configure media storage.	Domain tables exist in SQLite; verify migration status and basic model-backed API writes.
2 ? P0	Replace demo login/registration with real JWT APIs; load /users/me; provide organization/workspace onboarding.	Valid credentials required; API requests carry real JWT; new user has a usable tenant/workspace.
3 ? P0	Wire report list/create controls; define real report generation and export contract; produce valid PDF; implement XLSX if required.	Create report, generate PDF from actual report config/data, download it, and verify file content.
4 ? P0/P1	Wire dataset file selection/drop and upload; validate file type; parse JSON; implement preview and record storage/query strategy.	CSV/XLSX/JSON upload, persistence, preview, record retrieval, and clear error states work.
5 ? P1	Replace dashboard/analytics fixtures; implement missing routes and align API response shapes; derive metrics from persisted records.	Workspace/period filters change real server results; loading/empty/error states are visible.
6 ? P1	Enforce tenant membership/role checks on create/update and nested member/user endpoints.	Cross-tenant reads/writes are denied; roles gate admin/member operations.
7 ? P1	Pass workspace ID to AI query; build real authorized data context; implement insight/anomaly/forecast workflows.	AI responses reflect selected workspace; missing key/provider failures surfaced.
8 ? P2	Correct notification method/path; add audit writers; seed plans; clarify non-payment upgrade behavior; implement Celery schedule delivery and email configuration.	Notifications persist state, audit entries are written, plans are available, scheduled jobs execute.

F. Files likely to modify by feature

Feature	Likely files
Migrations/schema	backend//migrations/ (new initial migration packages); affected backend//models.py; backend/config/settings/base.py
Auth/onboarding	frontend/src/pages/auth/Login.tsx, Register.tsx, ForgotPassword.tsx; frontend/src/pages/app/Settings.tsx; frontend/src/services/api.ts, index.ts; frontend/src/store/index.ts; backend/authentication/views.py, serializers.py; backend/users/views.py; organization/workspace files.
Datasets	frontend/src/pages/app/Datasets.tsx; frontend/src/services/index.ts; backend/datasets/views.py, serializers.py, models.py, urls.py; backend/config/settings/base.py.
Dashboard/analytics	frontend/src/pages/app/DashboardHome.tsx, Analytics.tsx, DashboardList.tsx, DashboardBuilder.tsx, App.tsx; frontend/src/utils/mockData.ts; backend analytics/dashboard models, serializers, views, and urls.
AI/anomaly/forecast	frontend/src/pages/app/AIInsights.tsx; frontend/src/services/index.ts; backend/ai_insights/views.py, gemini_client.py, models.py, serializers.py, urls.py; analytics models/views/urls.
Reports/export/scheduling	frontend/src/pages/app/Reports.tsx; frontend/src/services/index.ts; backend/reports/views.py, serializers.py, models.py, urls.py; Celery config and new report task module.
Organization/security	frontend/src/pages/app/Organization.tsx, AdminPanel.tsx; frontend services/store; backend organizations/workspaces/users views, serializers, URLs, permissions.

Feature	Likely files
Notifications/audit	frontend/src/pages/app/Notifications.tsx, AuditLogs.tsx; frontend/src/components/layout/NotificationPanel.tsx, Topbar.tsx; frontend services; backend notifications/audit views, serializers, models, event writers.
Subscriptions/settings	frontend/src/pages/app/AdminPanel.tsx, Settings.tsx; frontend services; backend subscriptions/user/auth endpoints and models.

Audit boundary

This report reflects static source/configuration inspection and read-only database schema checks. It does not represent a live end-to-end verification of every endpoint. Django system check passed; project migrations are absent and no tests were run. No files were modified. The implementation plan is pending approval.